RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 02/06/2025	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 30 + 90 Min**
Course Title: STATISTICS FOR DATA ANALYTICS		**Course Code : IM241AT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	The binomial distribution models the number of _____ in a fixed number of independent trials.	1	2	2
2.	The area under the curve of a normal distribution represents the total probability and is always equal to _____.	1	1	2
3.	If the number of calls received per hour at a call center is Poisson distributed with a mean of 4, what is the probability that exactly 2 calls are received in an hour?	2	3	2
4.	Enumerate any two real-life situations where the continuous uniform distribution can be applied.	2	2	2
5.	What is the memoryless property of exponential distribution?	2	2	2
6.	For a normal distribution with $\mu = 70$, $\sigma = 8$, find $P(62 < X < 78)$.	2	2	2
	Part – B			
1	Past experience indicates that the ticket office at the Rex Theatre receives a mean rate of thirty calls per hour. Assuming that the number of calls have the Poisson distribution, find the probability that a) during a ten-minute coffee-break, no calls will be received b) during a twenty-minute coffee-break, at least two calls will be received c) during a half-an-hour lunch break, at most 3 calls will be received	10	3	2
2	A factory producing washing machines sends 1,000 machines each month to one of their customers. The manufacturer's past records show that 5% of their machines are defective. The customer decides to pick a sample of 20 machines and subject them to tests. a) What is the probability of the customer finding at most no defective in the sample? b) The customer tolerates at most 1 defective in the sample for accepting the consignment of 1,000 machines. What is the probability of the customer accepting the whole consignment? c) What is the mean number of defective machines in the sample?	10	3	2

3	The random variable X has a binomial distribution with n = 10 and p = 0.01. Determine the following probabilities. a) $P(X = 5)$ b) $P(X \leq 2)$ c) $P(X \geq 9)$ d) $P(3 \leq X < 5)$	10	3	2
4	Inter arrival times in minutes between successive customers in a hotel is exponentially distributed with a mean of 2 minutes. a) Compute the probability that inter arrival time between two customers is at most 2 minutes. b) What is the probability that at least 2 customers arrive in one minute? c) What is the probability of no customers arriving in a span of 3 minutes?	10	3	2
5	Suppose that a large conference room at a certain company can be reserved for no more than 4 hours. Both long and short meetings occur quite often. In fact, it can be assumed that the length X of a meeting is equally likely in the interval [1.5, 4]. a) What is the probability density function? b) What is the probability that any given meeting lasts at least 3 hours? c) Find the mean and variance of length X of the meetings.	10	3	2

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Date : 26th August 2024	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 120 Min**
Course Title: STATISTICS FOR DATA ANALYTICS		**Course Code : IM241AT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	A market researcher wants to determine if a new advertisement has changed customer preference between two products. What hypothesis test should be conducted?	2	1	4
2	Differentiate between Type I and Type II Errors.	2	1	4
3	A factory manager believes that the average production rate has decreased due to new machinery. What type of hypothesis test would you use?	2	2	4
4	What is the significance level (α) in hypothesis testing?	2	2	4
5	Differentiate between left tailed and right tailed test with an example.	2	1	4
	Part – B			
1	The mean water temperature downstream from a power plant cooling tower discharge pipe should be no more than 100°F. Past experience has indicated that the standard deviation of temperature is 2°F. The water temperature is measured on nine randomly chosen days, and the average temperature is found to be 98°F. a) Should the water temperature be judged acceptable with 0.05? b) What is the P-value for this test? c) What is the probability of accepting the null hypothesis at 0.05 if the water has a true mean temperature of 104°F?	10	4	4
2	A stenographer claims that she can type at the rate of 120 words per minute. Can we reject her claim on the basis of 100 trials in which she demonstrates a mean of 116 words with a standard deviation of 15 words. Use 5% level of significance.	10	3	4
3	A machine is designed to produce insulating washers for electrical devices of average thickness of 0.025 cm. A random sample of 10 washers was found to have average thickness of 0.024 cm with standard deviation of 0.002 cm. Test the significance of the deviation. The value of t for 9 degrees of freedom at 5% level is 2.262.	10	3	4
4	Calculate Karl Pearson's Correlation Coefficient between the marks in Statistics for Data Analytics and Operations Research obtained by 10 students. SDA: 10, 25, 13, 25, 22, 11, 12, 25, 21, 20 OR: 12, 22, 16, 15, 18, 18, 17, 23, 24, 17	10	3	3

<table>
<tr>
<td>5</td>
<td>A panel of judges P and Q graded seven debaters and independently awarded following marks.
<table>
<tr><td>Debaters</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr>
<tr><td>Marks by judge P</td><td>40</td><td>34</td><td>28</td><td>30</td><td>44</td><td>38</td><td>31</td></tr>
<tr><td>Marks by judge Q</td><td>32</td><td>39</td><td>26</td><td>30</td><td>38</td><td>34</td><td>28</td></tr>
</table>
An eighth debater was awarded 36 marks by judge P while judge Q was not present. If judge Q were also present, how many marks would you expect him to award to the eighth debater assuming the same degree of relationship exists between their judgement?</td>
<td>10</td>
<td>3</td>
<td>3</td>
</tr>
</table>

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DEPARTMENT OF
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Date : 5th September 2023	**Improvement Test**	**Max. Marks : 50**
Semester : IV	**UG**	**Duration : 90 Min**
Course Title: STATISTICS FOR DATA ANALYTICS		**Course Code : 21IM41**

Note:
1. Use of statistical tables permitted.

Sl. No	Questions	M	BT	CO
1.	An engineer who is studying the tensile strength of a steel alloy intended for use in golf club shafts knows that tensile strength is approximately normally distributed. A random sample of 12 specimens has a mean tensile strength of 3250 psi. (i) Test the hypothesis that mean strength is 3500 psi. Use 1% significance level. Standard deviation is known to be 60 psi. (ii) What is the smallest level of significance at which you would be willing to reject the null hypothesis in part (i)? (iii) Explain how you could answer the question in part (i) with a two-sided confidence interval on mean tensile strength. (iv) How will the test be performed if the standard deviation is not known?	10	3	3
2.	Inter arrival times in minutes between successive customers in a hotel is exponentially distributed with a mean of 2 minutes. (i) Compute the probability that inter arrival time between two customers is at most 2 minutes. (ii) What is the probability that at least 2 customers arrive in one minute? (iii) What is the mean-time till every 3rd customer arrives? (iv) What is the probability of no customers arriving in a span of 3 minutes?	10	3	3
3.	(i) What are the hypothesis statements and the statistic to be used in a two-sided test of variance of a population? What are the assumptions to use this statistic and what is the rejection region for this test? (ii) State the probability of a t-random variable with 25 degrees of freedom taking on any value between 2.06 and 3.45. (iii) What is the value x such that a chi-square random variable with 15 degrees of freedom takes on at the most a value of x with 95% likelihood?	10	2	4
4.	For the sample (5, 2, 3, 1, 0, 1, 3, 3) from a population, (i) Construct a 90% confidence interval on population mean if the population standard deviation is known to be 2. (ii) What should the sample size be in order to keep the width of the confidence interval to less than 2? (iii) Construct a 95% confidence interval on population mean if the population variance is unknown.	10	3	4

5.	An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find (i) the probability that a bulb burns between 778 and 834 hours. (ii) Probability that a bulb has a life of at least 1000 hours. (iii) What percent of bulbs have their life within $\pm 2\sigma$ from their own mean life, where σ is the standard deviation?	10	3	4

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Date : 2nd June 2025	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 30 + 90 Min**
Course Title: CAD/CAM & ROBOTICS		**Course Code : IM343AI**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	The physical boundary within which a robot operates is its _____.	1	1	3
2.	The robot joint that provides rotational motion is called a _____ joint.	1	2	4
3.	A _____ is a device at the end of a robot arm, designed to interact with the environment.	1	1	4
4.	Industrial robots typically follow a _____ cycle for repeated operations.	1	2	4
5.	_____ sensors gather information from the robot's external environment.	1	2	4
6.	Which type of programming is suitable for complex and repeatable tasks?	1	3	4
7.	What is teach pendant in robotics?	2	2	4
8.	What is a gripper in robotics?	2	1	3
	Part – B			
1	Describe the selection criteria and types of grippers used in robotics.	10	2	3
2	Explain the characteristics of industrial robots and how they impact manufacturing industries.	10	2	4
3	Explain the robot wrist and end-of-arm tooling (EOAT) with examples.	10	2	4
4	Describe in detail the role of sensors in robotics and how they improve robot performance.	10	1	4
5	Identify any four robot programming languages. Name the important requirement of programming languages.	10	1	4

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Date : 26th August 2024	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 30 + 90 Min**
Course Title: CAD/CAM & ROBOTICS		**Course Code : IM343AI**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	The word robot was introduced in 1920 in a play by _____.	1	1	4
2	Expand PUMA.	1	2	4
3	Cartesian Robot is also called as _____.	1	2	4
4	A device that is used to detect event or changes in the environment is called _____.	1	1	4
5	What is a work envelope of a robot?	2	2	4
6	Define zeroth law of Robotics.	2	1	4
7	What do you mean by task level programming?	2	2	4
	Part – B			
1	How do you specify a robot? Is robotics an automation? Discuss the different classification of robots.	10	1	4
2	Explain the selection criteria to select one among the different Robot.	10	2	4
3	What are the different types of sensors? Classify them. Sketch and explain the use of a velocity sensor.	10	2	4
4	Discuss the lead through programming technique used in robot programming.	10	1	4
5	Identify any four robot programming languages. Name the important requirement of programming languages.	10	2	4

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Date : 03/06/2025	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 30 + 90 Min**
Course Title: OPERATIONS RESEARCH		**Course Code : IM244AI**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	A _____ diagram is used to represent the logical sequence of project activities.	1	2	3
2.	The longest path through a project network is called the _____.	1	2	3
3.	Slack time in a PERT chart represents the amount of time an activity can be delayed without delaying the _____.	1	3	3
4.	_____ is the process of shortening the project duration by increasing resource allocation to critical activities.	1	1	3
5.	Software tools like MS Project and Primavera help in visualizing and managing _____.	1	2	3
6.	In a _____ game, the gain of one player is exactly the loss of the other.	1	3	4
7.	A strategy that always gives the best minimum gain is called a _____ strategy.	1	2	4
8.	If a game has a _____ point, then a saddle point solution exists using pure strategies.	1	3	4
9.	The _____ method is used to solve $2 \times n$ or $m \times 2$ games graphically.	1	3	4
10.	The rule of _____ helps to eliminate dominated strategies in a game matrix.	1	2	4
	Part – B			
1	Explain the concepts of Two-Person Zero-Sum Games and Pure Strategies. Describe how to solve a game with a saddle point using the payoff matrix. Illustrate with an example.	10	2	4

<table>
<tr>
<td>2</td>
<td>Listed in the table are the activities and sequencing necessary for a maintenance job on the heat exchangers in a refinery.
<table>
<tr><th>Activity</th><th>Description</th><th>Predecessor Activity</th></tr>
<tr><td>A</td><td>Dismantle pipe connections</td><td>–</td></tr>
<tr><td>B</td><td>Dismantle heater, closure, and floating front</td><td>A</td></tr>
<tr><td>C</td><td>Remove tube bundle</td><td>B</td></tr>
<tr><td>D</td><td>Clean bolts</td><td>B</td></tr>
<tr><td>E</td><td>Clean heater and floating head front</td><td>B</td></tr>
<tr><td>F</td><td>Clean tube bundle</td><td>C</td></tr>
<tr><td>G</td><td>Clean shell</td><td>C</td></tr>
<tr><td>H</td><td>Replace tube bundle</td><td>F, G</td></tr>
<tr><td>I</td><td>Prepare shell pressure test</td><td>D, E, H</td></tr>
<tr><td>J</td><td>Prepare tube pressure test and reassemble</td><td>I</td></tr>
</table>
Draw a network diagram of activities for the project. Find the critical path.</td>
<td>12</td><td>3</td><td>3</td>
</tr>
<tr>
<td>3</td>
<td>A small project involves 7 activities, and their time estimates are listed in the following table. Activities are identified by their beginning (i) and ending (j) node numbers.
<table>
<tr><th rowspan="2">Activity (i–j)</th><th colspan="3">Estimated Duration (weeks)</th></tr>
<tr><th>Optimistic</th><th>Most Likely</th><th>Pessimistic</th></tr>
<tr><td>1–2</td><td>1</td><td>1</td><td>7</td></tr>
<tr><td>1–3</td><td>1</td><td>4</td><td>7</td></tr>
<tr><td>1–4</td><td>2</td><td>2</td><td>8</td></tr>
<tr><td>2–5</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>3–5</td><td>2</td><td>5</td><td>14</td></tr>
<tr><td>4–6</td><td>2</td><td>5</td><td>8</td></tr>
<tr><td>5–6</td><td>3</td><td>6</td><td>15</td></tr>
</table>
a) Draw the network diagram of the activities in the project.
b) Find the expected duration and variance for each activity. What is the expected project length?
c) Calculate the variance and standard deviation of the project length. What is the probability that the project will be completed: (i) at least 4 weeks earlier than expected time; (ii) no more than 4 weeks later than expected time.</td>
<td>16</td><td>4</td><td>3</td>
</tr>
<tr>
<td>4</td>
<td>Use graphical method to find the optimal strategy for both the players and also value of the game for the following pay-off matrix:
<table>
<tr><th rowspan="2">Player A</th><th colspan="5">Player B</th></tr>
<tr><th>I</th><th>II</th><th>III</th><th>IV</th><th>V</th></tr>
<tr><td>I</td><td>3</td><td>0</td><td>6</td><td>−1</td><td>7</td></tr>
<tr><td>II</td><td>−1</td><td>5</td><td>−2</td><td>2</td><td>1</td></tr>
</table></td>
<td>12</td><td>4</td><td>4</td>
</tr>
</table>

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DEPARTMENT OF
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Date : 27th Aug 2024	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 30 + 90 Min**
Course Title: OPERATIONS RESEARCH		**Course Code : IM244AI**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	What is two person Zero-sum game?	2	1	1
2	For what value of λ, the game with following pay-off matrix is strictly determinable? Player A: A_1: λ, 6, 2; A_2: -1, λ, -7; A_3: -2, 4, λ (Player B: B_1, B_2, B_3)	2	2	2
3	Can the dominance method be used to solve all games? Give reason for your answer.	2	3	3
4	What is an unbalanced transportation problem? How to resolve it.	2	2	2
5	What is the purpose of finding an IBFS in the transportation problem?	2	2	2
	Part – B			
1	Describe the maximin principle of game theory. What do you understand by pure strategies and 'saddle point'.	10	2	2
2	Solve the following game for optimal strategy for both the players and value of the game by dominance property whose payoff matrix is: Player A: I: 4, 2, 0, 2, 1, 1; II: 4, 3, 1, 3, 2, 2; III: 4, 3, 7, -5, 1, 2; IV: 4, 3, 4, -1, 2, 2; V: 4, 3, 3, -2, 2, 2 (Player B: I, II, III, IV, V, VI)	10	3	3

<table>
<tr><td>3</td><td>Use the graphical method to find the optimal strategy for both the players and also value of the game for the following pay-off matrix:
<table>
<tr><td rowspan="2">Player A</td><td colspan="3">Player B</td></tr>
<tr><td>B1</td><td>B2</td><td>B3</td></tr>
<tr><td>A1</td><td>1</td><td>3</td><td>12</td></tr>
<tr><td>A2</td><td>8</td><td>6</td><td>2</td></tr>
</table></td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>4</td><td>Write a Linear Programming Mathematical Model for the transportation problem.</td><td>10</td><td>4</td><td>4</td></tr>
<tr><td>5</td><td>A firm has 3 factories 1, 2, and 3. There are four major warehouses situated at A, B, C, and D. Average daily product at 1, 2, 3 is 30, 40, and 50 units respectively. The average daily requirement of this product at A, B, C, and D is 35, 28, 32, 25 units respectively. The transportation cost (in Rs.) per unit of product from each factory to each warehouse is given below:
<table>
<tr><td rowspan="2">Factory</td><td colspan="4">Warehouse</td></tr>
<tr><td>A</td><td>B</td><td>C</td><td>D</td></tr>
<tr><td>1</td><td>6</td><td>8</td><td>8</td><td>10</td></tr>
<tr><td>2</td><td>5</td><td>11</td><td>9</td><td>7</td></tr>
<tr><td>3</td><td>8</td><td>9</td><td>7</td><td>13</td></tr>
</table>
Find the initial basic feasible solution by North west corner rule and row minima method.</td><td>10</td><td>2</td><td>2</td></tr>
</table>

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Date : 4th June 2025	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 30 + 90 Min**
Course Title: MARKETING MANAGEMENT		**Course Code : IM345AT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	What is the key difference between mobile advertising and mobile marketing?	2	1	1
2	Mention two key principles of User-Centered Design in website development.	2	2	2
3	How can website metrics help in optimizing a website for marketing effectiveness?	2	3	3
4	What does the AIDAS model stand for in conversion optimization?	2	1	1
5	State two key elements that visitors commonly expect to see on a well-optimized landing page.	2	2	3
	Part – B			
1	Explain the AIDAS model and its role in website conversion optimization. How can key elements of a website be optimized to increase landing page effectiveness?	10	3	1
2	Discuss what website visitors typically look for when they land on a business webpage. Suggest effective design and content strategies to meet those expectations.	10	3	2
3	Trace the evolution of digital analytics and its growing importance in delivering a seamless end-to-end customer experience.	10	1	1
4	Design a user-centered website layout for a new e-commerce startup targeting young urban professionals. How would you ensure optimal user experience and engagement? Include key design principles, optimization strategies, and website copy suggestions.	10	4	2
5	Assume you are launching a promotional campaign for a new product using mobile marketing. Prepare a detailed plan including mobile advertising channels, push notifications, app-based promotions, and integration with social media.	10	3	3

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Date : 28th August 2024	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 120 Min**
Course Title: MARKETING MANAGEMENT		**Course Code : IM345AT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	How does image optimization contribute to overall website performance?	2	2	3
2	List any four key metrics used to measure the success of a website?	2	2	3
3	How can push notifications be used in mobile marketing to enhance sales promotions?	2	2	3
4	Define mobile advertising in one sentence. How does it differ from mobile marketing?	2	2	4
5	How can user-friendly design impact the overall effectiveness of a website?	2	2	4
	Part – B			
1	Explain the AIDAS model in detail. Discuss how each stage plays a critical role in conversion optimization, using examples to illustrate how businesses can enhance customer journeys through this model.	10	2	3
2	A travel agency has created a landing page for a limited-time offer on holiday packages. Despite a high number of clicks on the ads leading to the landing page, the conversion rate is disappointingly low. Evaluate the possible reasons for the low conversion rate on the landing page. Suggest optimizations for key elements such as headlines, call-to-action buttons, and visuals that could enhance the effectiveness of the landing page and increase the conversion rate.	10	3	3
3	How do online applications serve as effective tools for mobile marketing? What are the key features that businesses should incorporate into their apps to ensure they are maximizing marketing potential?	10	2	3
4	A luxury cosmetics brand is planning to launch a flash sale exclusively through mobile devices. How would you design a mobile marketing campaign that not only drives awareness of the sale but also maximizes conversion rates during the limited time?	10	3	4
5	Discuss the relationship between design principles and website copy. How can visual design elements and well-crafted content work together to enhance user experience and achieve website objectives? Include examples of effective design and copy integration.	10	2	4

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Department of Mechanical Engineering

Date: June 2025	**Improvement CIE**	**Max. Marks: 30**
Semester: IV	**UG**	**Duration: 60 Min**
Course Title: Universal Human Values		**Course Code: HS248AT**

S No	Questions	M	BT	CO
	Quiz			
1	The participation of the human being in ensuring the role of physical facility in nurture, protection and providing means for the body is called its _____. (a) Utility value (b) Artistic value (c) Activity (d) Energy	1	2	1
2	When something is active or has activity, it is called as _____. (a) space (b) unit (c) energy (d) value	1	3	2
3	The following are the characteristics of space except _____. (a) Unlimited (b) No activity (c) Energy in equilibrium (d) Self-organized	1	1	1
4	The natural characteristic of material order is _____. (a) Composition (b) Decomposition (c) Composition/Decomposition (d) Nurture/worsen	1	1	1
5	Conformance of material order is named as _____. (a) constitution conformance (b) Seed conformance (c) Breed conformance (d) sanskaar conformance	1	3	3
	Test			
1	Explain the classification of units into Four Orders and the harmony among them with the help of a schematic diagram.	6	2	1
2	What do you mean by ‘conformance’? Explain the conformance in the four orders.	6	4	2
3	Explain how there is recyclability and self-regulation in nature.	6	3	3
4	Elaborate the terms ‘units’ and ‘space’ and mention their characteristics.	7	4	4

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Date : 28 August 2024	**Improvement CIE – Quiz and Test**	**Max. Marks : 25**
Semester : IV	**UG**	**Duration : 60 Min**
Course Title: UNIVERSAL HUMAN VALUES		**Course Code : HS248AT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
Part – A				**(5 Marks)**
1	What is the fundamental principle of harmony in nature and existence? a) Competition b) Coexistence c) Dominance d) Isolation	1	1	3
2	What is the role of human beings in maintaining harmony in nature? a) Exploiting natural resources b) Preserving and nurturing nature c) Dominating other species d) Isolating from the environment	1	2	2
3	Which of the following best describes the Physical Order? a) Includes only human-made objects b) Comprises natural elements like soil, water, air, and minerals c) Refers exclusively to living beings d) Focuses on mental and spiritual aspects	1	1	2
4	What distinguishes the Human Order from the other three orders? a) Physical strength b) Intelligence and moral values c) Ability to photosynthesize d) Instinctual behaviour only	1	1	4
5	In the context of coexistence, what is the meaning of ‘prosperity’? a) Material wealth accumulation b) Holistic well-being and happiness c) Physical strength and power d) Technological superiority	1	1	3
Part – B				
1	Existence is co-existence of mutually interacting units in all-pervasive space. Explain.	5	2	3
2	What is the natural characteristics (swabhava) of human order? Explain.	5	1	2
3	What are the four orders of nature? Briefly explain them.	5	2	4
4	How can we say that ‘nature is Self-Organized and in space Self-Organization Is Available.’	5	2	4

5	Explain the difference and similarities between Pranic order and animal order. What is the relation between the two orders?	5	1	2

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Department of Mechanical Engineering

Date: 03rd June 2025	**CIE – Improvement**	**Max. Marks: 10 + 50**
Semester: IV	**UG**	**Duration: 2 Hrs**
Course Title: Materials Science for Engineers		**Course Code: ME242TB**

S No	Questions	M	BT	CO
	Part A			
1	Mention the purpose of baking in electronic devices.	1	1	3
2	_____ is a process used to grow a thin layer of silicon dioxide (SiO_2) on the surface of silicon wafers by exposing them to an oxidizing atmosphere at elevated temperatures.	1	1	3
3	Highlight any two advantages of Rapid Thermal Processing.	2	2	3
4	Name the heat treatment process variables.	2	1	3
5	Define nanomaterials.	2	1	4
6	Give two examples for top-down approaches of nanomaterials synthesis.	2	1	4
	Part B			
1(i)	Discuss the need of post-processing heat treatment methods for electronic devices.	5	2	3
1(ii)	Briefly explain the steps to be followed in Solder Reflow process.	5	3	3
2(i)	Differentiate Annealing and Normalizing.	4	2	3
2(ii)	With a neat sketch describe induction hardening heat treatment process.	6	3	3
3	Define heat treatment. List and Explain the defects in the heat treatment process.	10	3	3
4	Illustrate with a neat sketch the Chemical vapour deposition technique for the synthesis of nanomaterials.	10	3	4
5	Discuss the working principle of Scanning Electron Microscope characterization technique of nanostructures using a neat labelled diagram.	10	2	4

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Department of Mechanical Engineering

Date: August 2024	**CIE – Improvement**	**Max. Marks: 50 + 10**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Material Science for Engineers		**Course Code: ME242AT**

S No	Questions	M	BT	CO
	Part A			
1	Name a critical electronics manufacturing process primarily used in surface-mount technology.	1	1	3
2	Write two key advantages of RTP in Semiconductor manufacturing.	2	1	3
3	What are the two basic purposes of tempering.	2	1	3
4	The crystal structure of martensite is _____.	1	1	3
5	Mention the cause of warpage in heat treatment of mechanical structure.	1	1	3
6	Mention a property of a material which significantly defer from bulk material to nanomaterial.	1	1	4
7	CVD technique used for synthesis of a nanomaterial is a _____ type of approach.	1	1	4
8	How nanoFRP is different from FRP.	1	1	4
	Part B			
1a	Describe the steps followed in thermal oxidation process for the post processing heat treatment of electronic devices.	5	3	3
1b	Highlight the factors affecting thermal oxidation process of electronic devices.	5	2	3
2	Highlight the advantages, disadvantages and microstructures of following heat treatment processes for eutectoid steels: i) Annealing, ii) Normalizing and iii) hardening.	10	2	3
3a	Describe the process of construction of Time Temperature Transformation (TTT) curves of eutectoid steel.	5	3	3
3b	Explain the induction hardening process with a neat sketch.	5	3	3
4a	Define and classify nanomaterials with examples.	5	2	4
4b	Illustrate the synthesis of nanopowder using sol-gel method.	5	3	4
5a	Describe the properties and applications of carbon nanotubes and nano fabrics.	5	3	4
5b	Illustrate the working principle of Scanning Electron Microscope (SEM) for the characterisation of nanocomposites.	5	3	4

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Department of Biotechnology

Date: 3rd June 2025	**Improvement Test**	**Max. Marks: 50 + 10**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT242AT**

S No	Questions (Quiz)	M	BT	CO
1	List out with two examples smart packing methods.	1	1	3
2	What is GLP? Give the outcomes of GMP in food industry.	1	1	3
3	Why antibiotics are used in food producing animals.	1	1	3
4	What is blanching? Give two advantages of food processing.	1	1	4
5	List out the health policies in India.	1	1	4
6	What are ethics in food industry?	1	1	4
7	What is food production and economics?	2	1	4
8	What are the requirements of Food packing material. Give any two examples of food packing material.	2	2	3

S No	Questions (Test)	M	BT	CO
1	Give an account of various principles and factors of Research ethics.	10	3	4
2	Illustrate on various instruments used in food testing labs for analysis with applications.	10	3	3
3	Elaborate on the principles of clinical ethics with examples; add a note on physician obligations.	10	2	4
4	Elaborate on novel packaging methods with suitable examples.	10	2	3
5	Classify and explain the importance of food additives. Also add a note on the permissible limits for few food products.	10	1	3

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Department of Biotechnology

Date: 27th August 2024	**Improvement Test**	**Max. Marks: 50 + 10**
Semester: IV	**UG**	**Duration: 110 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT242AT**

S No	Questions (Quiz)	M	BT	CO
1	Explain with two examples about food allergens.	1	1	3
2	What is Research ethics?	1	1	3
3	What are the advantages of preservation of food?	1	1	3
4	What are flavouring agents used in food? Give examples.	1	1	4
5	List out the health policies in India.	2	1	4
6	What is ethics in food industry?	2	1	4
7	List out 4 methods used for food analysis.	2	1	4

S No	Questions (Test)	M	BT	CO
1	Suggest and explain on various strategies involved in food preservation.	10	2	3
2	Illustrate and classify on various food packaging methods used in food industry.	10	3	4
3	"Meat and meat products derived from livestock treated with antibiotics and hormones can cause adverse effects on consumer's health" justify the statement.	10	3	3
4	Elaborate on smart packaging methods with suitable examples. Add a note on packaging materials.	10	4	4
5	Classify and explain the importance of food additives. Also add the permissible limits of few food additives.	10	2	3

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Department of Civil Engineering

Date: June 2025	**Improvement CIE (Quiz + Test)**	**Max. Marks: 50 + 10**
Semester: IV	**UG**	**Duration: 90 Min + 20 Min**
Course Title: Environment and Sustainability (Basket Course)		**Course Code: CV242AT**

S No	Questions	M	BT	CO
	Part A			
1.	List the main focus of environmental economics?	1	1	1
2.	What does GDP stand for?	1	2	1
3.	Name one key difference between linear and cyclical resource management systems.	1	3	2
4.	What is the goal of a circular economy?	1	2	2
5.	What does ISO 14000 series focus on?	1	2	1
6.	Describe the significance of carbon credits used for?	1	1	3
7.	What is the meaning of the “R” in zero waste and R concept?	1	1	1
8.	Give an example for cyclical systems.	1	2	4
9.	What is a green building?	1	1	3
10.	Define Environmental Impact Assessment (EIA)?	1	2	4
	Part B			
1.	Explain the concept of sustainability. Discuss the challenges in achieving sustainable development.	10	2	2
2.	Differentiate between linear and cyclical resource management systems. How can cyclical systems contribute to sustainability?	10	2	3
3.	Describe the Sustainable Development Goals (SDGs). Explain any three goals in detail with their targets and indicators.	10	2	4
4.	What is Environmental Impact Assessment (EIA)? Explain its process and importance in sustainable development.	10	2	1
5.	Explain the role of green technology and industrial ecology in achieving sustainable development.	10	2	4

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Department of Civil Engineering

Date: 27/08/2024	**Improvement CIE**	**Max. Marks: 10Q + 50T**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Environment and Sustainability		**Course Code: CV242TA**

S No	Improvement Quiz – Questions	M	BT	CO
1.1	Mention any two major causes for climate changes.	2	1	3
1.2	List the benefits of circular economy?	2	1	3
1.3	Discuss energy efficiency, and why is it important in reducing environmental impact?	2	2	3
1.4	List the advantages of Green building.	2	1	4
1.5	Discuss the challenges, when trying to implement the 5Rs?	2	2	4

S No	Improvement Test – Questions	M	BT	CO
1.	Explain the key objectives of conducting an Environmental Impact Assessment (EIA).	10	2	3
2.	Discuss the role of Life Cycle Assessment (LCA) in calculating the carbon footprint of a product.	10	2	3
3.	Explain the principles of Corporate Social Responsibility.	10	2	4
4.	Discuss the history and evolution of Corporate Social Responsibility (CSR).	12	2	4
5.	Discuss the benefits of corporate philanthropy.	08	2	4